

Basic Of Internet

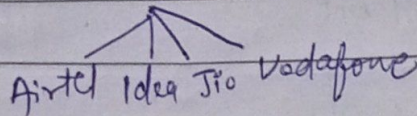
Lec 7

OSI Model

Date: 24/9/24
Page: _____

→ Subnet masking (in easy way)

mobile : sim



Idea : 99 _____ 10 no.s (0-9) Phone No.

Airtel : 86 _____ $10^8 = \text{phone no.s}$
95 _____

Vodafone : 932 _____ 10^7 sim
less $\downarrow$ sim required generate

more no.s needed → 6 _____
also started in India

+91 → India country code

box same number in diff country can be verified with country code that which person is being called.

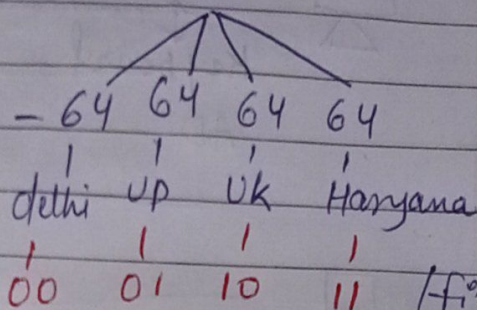
10^{10} → 1000 crore can be made (phone no.s)
150 crore population in India.

If more needed we can make 12 digit phone no.

Idea

198.67.28.0 (Class C)
N/w ID host ID
28 = 256

8 bit =



00 00 00 00 = 198.67.28.0

00 11 11 11 = 198.67.28.63 } delhi

198.67.28.0 / 26

N/w ID (26 bits)

32 - 26 = 6 bit (host ID)

2⁶ = 64

↳ host for delhi

Same for other three.

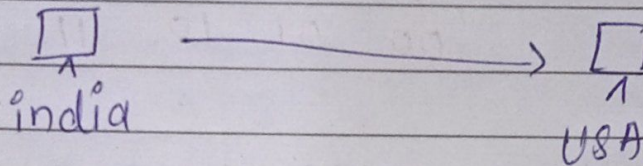
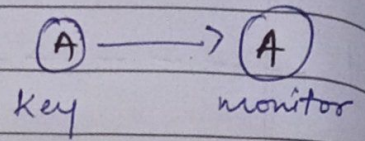
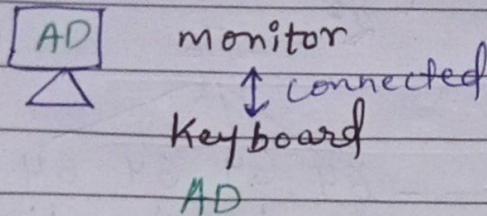
To distribute in

2 parts
4 parts
8 parts
reserve

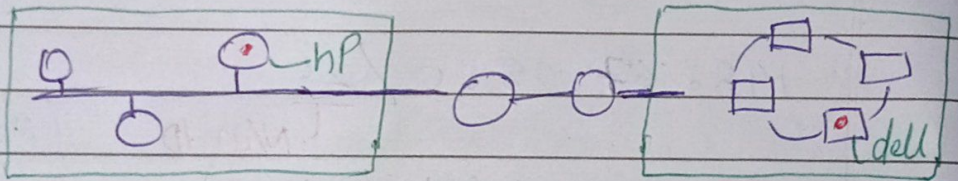
1 bit, 2 bits,
3 bits
respectively.

→ OSI Model

Open System Interconnection



Computer network



How data transferred over the n/w

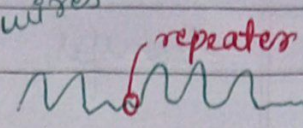
Protocol (common)
↳ OSI

Application layer
Presentation layer
Session layer
Transport layer
Network layer
Data link layer
Physical layer

7 layers ↓
client → 7 layers ↑
server
transmission

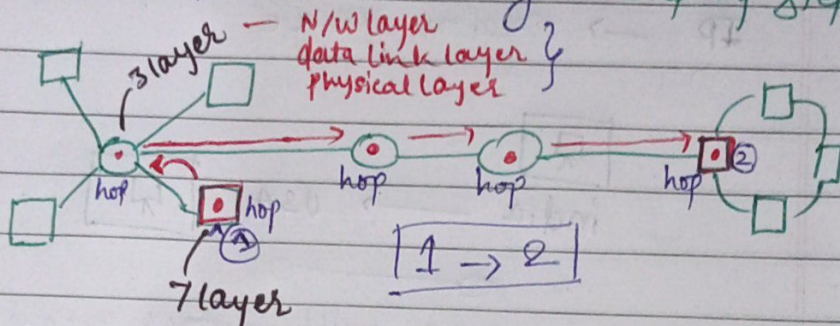
7 layers of OSI

1. Physical Layer

- Cables & connector - wires & switch/hub
- repeater -  so on
- data rate control - control over 3mbps
10mbps
- encoding - analog $\rightarrow$ digital
- physical topology - ring / bus / etc.

2. Data Link Layer

- hop to hop delivery - step by step



- giving physical address

media access control $\rightarrow$ MAC address

every device has unique mac address but not restricted that only one.

To check mac add:-

mac : networksetup - listall hardware reports? } cmd
windows : getmac /v /fo list

NIC $\rightarrow$ network interface card

$\downarrow$
in system to connect to internet

all the ways we can connect to internet
that many MAC address in system

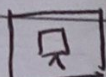
(wifi, ethernet, bluetooth)
- in my system (3)

In DLI mac address is used to transfer data

↳ bcoz double verification (IP + MAC)

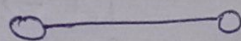
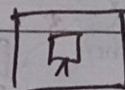
SIP	Data	DIP
	Sender MAC	Recd. MAC

IP → region specific



india

→ USA



how will routers know you are in USA -

also MAC add. can be changed

that's why MAC

is not used alone

IP is also necessary

MAC address is used locally or near about devices. (in LAN)

↳ only MAC

add. is enough

NIC

device → wifi - nic - MAC add.
 cable - nic - MAC add.
 usb - nic - MAC add.

↳ diff.